

MCA — CURRICULUM & SYLLABUS

Academic Year 2026–27

FY MCA 2026 Pattern

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FIRST YEAR MCA — SEMESTER I

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1. DATA STRUCTURES AND ALGORITHMS

Unit I — Fundamentals of Algorithms and Linear Data Structures

Unit II — Linked Lists

Unit III — Trees

Unit IV — Graphs & Hashing

Unit V — Searching & Sorting

2. MATHEMATICAL FOUNDATIONS

Unit I — Logic, Set and Relation

Unit II — Probability, Permutations and Combinations

Unit III — Random Variables and Probability Distributions

Unit IV — Graph Theory and Trees

3. SOFTWARE DEVELOPMENT METHODOLOGIES

Unit I — Software Engineering Fundamentals and Process Models

Unit II — Software Requirements and Design Methodologies

Unit III — Agile Software Development Methodologies

Unit IV — Software Project Planning, Estimation and Maintenance

4. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Unit I — Introduction to Artificial Intelligence

Unit II — Knowledge Representation in AI

Unit III — Introduction to Machine Learning

Unit IV — Mathematical & Statistical Foundation for Machine Learning

5. PYTHON PROGRAMMING LABORATORY

Python Programming Lab

6. ELECTIVE — CHOOSE ONE

OPTION A — DEEP LEARNING

Unit I — Introduction to Deep Learning

Unit II — Fundamentals of Neural Network

Unit III — Convolutional Neural Networks

Unit IV — Recurrent Neural Networks

Unit V — Autoencoders

Unit VI — Applications of Deep Learning

OR

OPTION B — UI/UX DESIGN

Unit I — UI/UX Foundations and User-Centered Design

Unit II — Information Architecture and Visual Design

Unit III — UI Design, Prototyping and Design Systems

Unit IV — Usability, Accessibility and Emerging Trends

OR

OPTION C — DATA MINING AND BUSINESS INTELLIGENCE

Unit I — Introduction of Data Mining and Data Warehouse

Unit II — Data Preprocessing

Unit III — Data Mining Techniques

Unit IV — Business Intelligence and Data Visualization

7. INTRODUCTION TO CYBER SECURITY

Unit I — Introduction to Cyber Security

Unit II — Cyber-attacks and Cybercrimes

Unit III — Computer Forensics

Unit IV — Cryptography and Emerging Areas in Cyber Security

8. HUMAN RIGHTS — I

Unit I — Human Values and Human Rights

Unit II — Rights and Duties

Unit III — Legal Instruments for Human Rights

Unit IV — United Nations and Human Rights

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FIRST YEAR MCA — SEMESTER II

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1. OPERATING SYSTEMS AND COMPUTER NETWORKS

Unit I — Introduction to Operating Systems & Process Management

Unit II — Memory Management & Storage Management

Unit III — Networking Basics & Data Link Layer

Unit IV — Network Layer & Transport Layer

Unit V — Application Layer & Network Security

2. JAVA PROGRAMMING

Unit I — Java Fundamentals and Object-Oriented Programming

Unit II — Exception Handling and Collections Framework

Unit III — Multithreading and File Handling

Unit IV — JDBC and Modern Java Development

3. DATABASE MANAGEMENT SYSTEM

Unit I — Fundamentals of DBMS

Unit II — Database Design Using ERD

Unit III — SQL and PL/SQL

Unit IV — NoSQL Databases

4. INTERNET OF THINGS

Unit I — Introduction to Internet of Things

Unit II — IoT Communication Technologies and Protocols

Unit III — IoT Hardware, Platforms and Embedded Programming

Unit IV — IoT Security and Industrial IoT

5. ELECTIVE — CHOOSE ONE

OPTION A — GENERATIVE AI

Generative AI

OR

OPTION B — MOBILE APPLICATION DEVELOPMENT

Mobile Application Development

OR

OPTION C — BIG DATA ANALYTICS

Unit I — Introduction to Big Data Analytics

Unit II — Big Data Management

Unit III — Big Data Processing and Lakehouse

Unit IV — Big Data in Business Context

6. OPEN ELECTIVE — CHOOSE ONE

OPTION A — E-BUSINESS & RETAILING

Unit I — Fundamentals of E-Business and E-Commerce

Unit II — Digital Marketing and Emerging Trends in E-Business

Unit III — Digital Payment Systems and E-Commerce Platforms

Unit IV — Retail Management and Analytics

OR

OPTION B — TECHNICAL CONTENT WRITING

Technical Content Writing

7. RESEARCH METHODOLOGY

Research Methodology

8. HUMAN RIGHTS — II

Unit I — Indian Context on Human Rights

Unit II — Enforcement Mechanism for Human Rights

Unit III — Legal Redress on Human Rights Violation

Unit IV — Role of Advocacy Groups in Protecting Human Rights

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SECOND YEAR MCA — SEMESTER III

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1. SOFTWARE TESTING AND QUALITY ASSURANCE

Unit I — Introduction to Software Quality Assurance

Unit II — Basics of Software Testing

Unit III — Software Testing Techniques

Unit IV — Software Testing Strategies

Unit V — Software Testing Tools — An Overview

Unit VI — Selenium Tool

2. DATA SCIENCE

Unit I — Introduction to Data Science

Unit II — Data Warehouse and Data Preprocessing

Unit III — Classification and Regression

Unit IV — Clustering

Unit V — Association Rule Mining

Unit VI — Data Visualization

3. WEB PROGRAMMING

Unit I — Scripting Language — HTML

Unit II — Client-Side Technology — JavaScript

Unit III — Bootstrap & WordPress

Unit IV — AngularJS & jQuery

Unit V — Server-Side Technology — PHP

Unit VI — Web Services

4. ELECTIVE III — CHOOSE ONE

OPTION A — DEEP LEARNING

Unit I — Introduction to Deep Learning

Unit II — Fundamentals of Neural Network

Unit III — Convolutional Neural Networks (CNN)

Unit IV — Recurrent Neural Networks (RNN)

Unit V — Autoencoders

Unit VI — Applications of Deep Learning

OR

OPTION B — CLOUD COMPUTING

Unit I — Introduction to Cloud Computing

Unit II — Cloud Computing Services

Unit III — Virtualization

Unit IV — Cloud Security

Unit V — Cloud Platforms and Cloud Applications

Unit VI — Future of Cloud Computing

OR

OPTION C — INFORMATION RETRIEVAL

Unit I — Introduction

Unit II — Dictionaries and Query Processing

Unit III — Text Classification & Text Clustering

Unit IV — Language Models for Information Retrieval

Unit V — Web Retrieval and Web Crawling

Unit VI — IR Applications

5. DATA SCIENCE LABORATORY

Main Practical Topics:

Logistic Regression

Decision Tree

Naive Bayes

Linear Regression

K-Means Clustering

FP-Tree / FP-Growth

Apriori

Association Rule Mining

Data Visualization

6. WEB PROGRAMMING LABORATORY

Practical work based on:

HTML

CSS

JavaScript

Bootstrap

WordPress

AngularJS

jQuery

PHP

Web Services

7. LABORATORY PRACTICE — III

Practical subject based on the Semester III elective/course topics.

8. RESEARCH PROJECT

Research Project

9. SKILL DEVELOPMENT — I

Skill Development — I

10. CONSTITUTION OF INDIA

Constitution of India

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SECOND YEAR MCA — SEMESTER IV

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1. ELECTIVE IV — NATURAL LANGUAGE PROCESSING & GENERATIVE AI

Unit I — Introduction to Natural Language Processing (NLP)

Unit II — Text Pre-processing

Unit III — Word Level Analysis

Unit IV — Language Modeling

Unit V — Syntax and Semantic Analysis

Unit VI — Generative AI

2. ELECTIVE IV — BLOCKCHAIN

Unit I — Mathematical Foundation for Blockchain

Unit II — Introduction to Blockchain

Unit III — Consensus in Blockchain

Unit IV — Cryptocurrency

Unit V — Blockchain Ethereum Platform using Solidity

Unit VI — Blockchain Case Studies

3. ELECTIVE IV — ADVANCED BUSINESS INTELLIGENCE AND ANALYTICS

Unit I — Overview of Business Intelligence and Business Analytics

Unit II — BI as a Business Support

Unit III — Advanced Data Analytics

Unit IV — Data Analysis Dashboards

Unit V — Data Reporting and Visualization

Unit VI — Project Management

4. ELECTIVE V — CHOOSE ONE

OPTION A — AI: CONSTRAINT SATISFACTION

NPTEL Course

OR

OPTION B — FOUNDATION OF CLOUD IoT EDGE ML

NPTEL Course

OR

OPTION C — EFFECTIVE WRITING

NPTEL Course

5. INDUSTRIAL TRAINING / RESEARCH PROJECT

Project Components:

1. Project Problem Statement
2. Aim, Objectives & Scope
3. Requirements
4. Requirement Analysis
5. UML / ER Diagrams
6. System Architecture & Design
7. Algorithms / Methods
8. Implementation
9. Test Cases
10. Optimization
11. Project Report
12. Paper Publication

6. SKILL DEVELOPMENT — II

Course: Enhancing Soft Skills and Personality

Mode: SWAYAM / NPTEL

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MCA 2026–27 — OVERALL TECHNOLOGY AREAS

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Programming:

Python, Java, JavaScript, PHP, SQL, PL/SQL, Solidity

AI / ML:

Artificial Intelligence, Machine Learning, Deep Learning,

Natural Language Processing, Generative AI

Data:

Data Science, Data Mining, Big Data Analytics,

Business Intelligence, Data Visualization

Development:

Web Development, Mobile Application Development,

IoT, Cloud Computing

Core Computer Science:

Data Structures & Algorithms, Operating Systems,

Computer Networks, DBMS, Software Engineering

Security:

Cyber Security, Cryptography, Computer Forensics,

Blockchain

Research & Professional Development:

Research Methodology, Research Project,

Industrial Training, Paper Publication,

Technical Content Writing, UI/UX,

Soft Skills & Personality Development

ATTENDANCE

If 75% attendance is mandatory, an emergency/medical situation may be considered if you inform your GFM (Guardian Faculty Member) and provide proper supporting documents.

Typical documents could include:

- Medical certificate
- Hospital/doctor documents
- Other official proof of the emergency

The final decision on whether the absence is accepted would depend on the college/department rules and approval from the concerned faculty/authority.

MCA 2026 – Grading & Academic Rules

- The MCA 2026 batch follows the 2026 Pattern applicable from Academic Year 2026–27.
- Assessment is based on:
 - CIE – Continuous Internal Evaluation
 - SEE – Semester End Examination
- Students must satisfy the minimum attendance requirement. As informed by the department, 75% attendance is mandatory.
- In case of an emergency or medical issue, the student should inform the GFM (Guardian Faculty Member) and submit proper supporting documents such as medical certificates or other valid proof.
- Approval of attendance consideration is subject to the college/department rules and the concerned authority.
- The grading system, passing criteria, SGPA/CGPA calculation and ATKT/backlog rules are as per the applicable MCA 2026 academic regulations.

